

**EVALUATING FINANCE-SPECIFIC LARGE LANGUAGE MODELS: A MULTI-
CRITERIA DECISION ANALYSIS (MCDA)**

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ABSTRACT

The finance industry utilises large language models (LLMs) to enhance analytical precision, regulatory compliance, and operational performance. The adoption process remains challenging as decision makers need to assess performance against governance requirements and operational feasibility. Existing assessments of finance-oriented LLMs are fragmented and often fail to address the operational challenges that affect real-world implementation.

The research employed a Multi-Criteria Decision Analysis (MCDA) framework to assess three fundamental aspects of open-source LLMs, namely: **Task Performance, Operational Capability, and Licence & Regulation**. The analysis incorporated non-compensatory constraints that focused on open-source access, documentation, and English language availability to ensure practical implementation. The research data originated from published benchmark tests, model documentation, and financial AI literature. The additive value model produced rankings, while sensitivity and scenario analyses evaluated how different stakeholder priorities influenced the results.

The findings demonstrated that different models require different trade-offs. The performance-based assessments consistently ranked FinLLaMA-7B-Instruct as the top model because it excelled in accuracy-driven tasks. FinMA-7B and Mistral-7B-Instruct-v0.1 achieved better results in compliance-led scenarios as a result of their permissive licensing, though they showed weaker benchmark performance. FinTral-Instruct's strong domain-specific fine-tuning was undermined by ambiguous licensing terms. The results indicated that benchmark performance did not serve as a sufficient indicator for deployment readiness since governance and operational requirements maintained equal weight.

The study presents a systematic evaluation framework for LLM assessment, which enables both academic and practical users to select models transparently and defend their choices. Future research should focus on developing standardised benchmarks, incorporating measures such as hallucination risk and environmental impact, and validating results through real-world deployment.

1. INTRODUCTION

In recent years, the finance industry heavily depends on data-driven systems which have established large language models (LLMs) as essential financial industry innovation. The processing of complex financial texts along with information synthesis and decision support capabilities of LLMs has elevated expectations for improved efficiency and accuracy in analytical, compliance, and operational workflows (Srivastava et al., 2024). The adoption process remains complex, as the finance industry faces distinctive challenges, including documents with dense numerical data, strict regulatory requirements, and the need for both transparency and auditability from models (Board of Governors of the Federal Reserve System, 2011; European Commission, 2024). General-purpose models prove insufficient for direct use because domain-specific LLMs remain in development, and their evaluation results are inconsistent (Yang et al., 2023).

The main challenge arises from the different assessment methods used by researchers. Researchers publish models with specific benchmarks that demonstrate their strengths, but these benchmarks do not provide standard comparison points between different alternatives (Bhatia et al., 2024). The current emphasis on specific performance metrics fails to consider essential factors, including licensing clarity, operational feasibility, and regulatory compliance that financial institutions need to address before deployment (Kang & Liu, 2023; Xie et al., 2024). These factors are essential for organizations because they establish the conditions for responsible and sustainable model deployment.

In light of these challenges, the research implements Multi-Criteria Decision Analysis (MCDA) to evaluate major open-source financial LLMs. The decision-making method of MCDA proves effective for this decision as it organizes complex evaluations between competing elements (Belton & Stewart, 2002). The framework enables complete transparency during performance assessment as it combines benchmark scores with governance requirements and operational feasibility. The study employs sensitivity analysis to validate its results by checking how different stakeholder priorities affect the ranking system, which produces recommendations that remain stable across various perspectives.

The research offers a valuable contribution by demonstrating how MCDA tools help organisations uncover hidden trade-offs that only emerge when multiple evaluation criteria are considered simultaneously (Keeney & Raiffa, 1993). The dissertation provides both theoretical

insights into model assessment techniques and practical guidance for organisations evaluating the benefits and limitations of implementing LLMs in financial operations.

2. LITERATURE REVIEW

2.1 Orientation and Scope

Over the past few years, the finance industry has been increasingly exploring domain-specialised LLMs, where general-purpose LLM models underperform due to the complexity in financial decision-making (Srivastava *et al.*, 2024). In the past two years, the research in finance-specific LLMs has been consolidated into three complementary strategies: (i) continued pre-training on finance corpora, (ii) instruction tuning with domain tasks, and (iii) multimodal alignment for tables and charts (Bhatia *et al.*, 2024). The main aim of the LLM models is to increase accuracy on finance tasks while ensuring transparency (Yang *et al.*, 2023).

Beyond accuracy, many finance tasks involve long inputs such as earnings-call transcripts, SEC filings, and table-heavy disclosures that exceed the context window. Hence, the context window is treated as a distinct operational factor, separate from the accuracy metrics, to capture its impact on feasibility (Mukherjee *et al.*, 2022). The LLM models explored in this dissertation are **FinMA-7B** (FinMA), **FinLLaMA-7B-Instruct** (FinLLaMA), **Llama-2-13B-Chat** (Llama-2), **Mistral-7B-Instruct-v0.1** (Mistral), and **FinTral-Instruct** (FinTral).

Previous research has rarely integrated finance-specific benchmarks with regulatory and operational factors within a unified framework. This dissertation aims to connect these gaps by using MCDA to integrate these dimensions.

2.2 Selection and Characteristics of Finance LLMs

This study evaluates five open-source LLM models that vary in architecture, training scope, and domain adoption. A few financial models such as Bloomberg, FinGPT, and InvestLM were excluded from the study. BloombergGPT is excluded as the LLM model is closed source, hence restricting access to training data and weights. This prevents transparent replication or independent evaluation (Wu *et al.*, 2023). In contrast, FinGPT and InvestLM are open-source models, but they are excluded due to scope and compatibility. FinGPT is designed as a continual learning framework that adapts smaller backbone models rather than a stable benchmark-based model suitable for MCDA (Yang *et al.*, 2023). Similarly, InvestLM is limited in scope as its primary aim is investment-specific tasks like portfolio reasoning and hence lacks broader coverage of natural language processing (NLP) benchmarks (Yang, Tang & Tam, 2023). Therefore, regarding real-world deployment of models in the finance industry, the

following models have been selected to capture trade-offs between domain specialisation, instruction following, licensing, and operational feasibility.

FinMA is the first publicly released financial LLM fine-tuned with instructions. The model is built on LLaMA and incorporates a 128K instruction dataset covering six finance natural language processing (NLP) tasks and two prediction tasks, which include credit-scoring and stock movement (Xie et al., 2023). While the model depicts strong instruction-following ability and verifiability, its limitations include weak quantitative reasoning and context length constraints, which are consequences of the original LLaMA architecture.

FinTral is one of the most recent model suites based on Mistral and applies domain-specific continued pre-training, instruction tuning and Direct Preference Optimisation (DPO) alignment. The model is trained using FinSet, which is a comprehensive benchmark suite. FinSet covers over 25 datasets across eight to nine financial task categories (Bhatia et al., 2024). FinTral was engineered to handle finance-specific tasks in a parameter-efficient manner to ensure clarity and consistent performance. The model does not leverage external tools or retrieval during inference but achieves competitive performance on zero and few-shot financial benchmarks, which makes it a strong candidate for evaluation.

The Open-FinLLMs family includes FinLLaMA which processes 52 billion tokens from financial domains that include SEC filings, macroeconomic indicators, and historical market data. The model receives 573k finance-specific prompts for instruction tuning to improve its performance in various financial NLP tasks (Xie et al., 2024). The Open-FinLLMs benchmark evaluates models through 14 financial tasks and more than 30 datasets to establish FinLLaMA as a reliable open-source solution for financial domain applications. The study examines FinLLaMA exclusively because it matches the requirements of text-based tasks.

The research incorporates two general-purpose baselines which demonstrate high performance: Llama-2 and Mistral. The Meta-developed Llama-2 model receives RLHF training and instruction-tuning across multiple conversational and analytical tasks (Touvron et al., 2023a). The model demonstrates strong performance in general but requires training on financial data and shows known limitations in quantitative and visual reasoning tasks.

Mistral-7B from Mistral AI achieves fast and efficient inference through its dense transformer architecture. The model is established under Apache 2.0 licensing terms and businesses can use it for commercial purposes. The model functions as a reference point for applications that need

low-latency performance. Similar to Llama-2, the model is not trained on a financial dataset and lacks optimization for regulatory and numeric financial reasoning (Jiang et al., 2023).

Collectively, these models represent the broad range of financial LLM, from domain-specific models to generalised models, making them suited for the criteria-based comparison carried out in this dissertation using the MCDA framework.

Model	Base Architecture	Training Scope / Data	Licence Type	Strengths	Limitations
FinMA	LLaMA-7B	128K finance instructions; 6 NLP + 2 predictive tasks (Xie et al., 2023)	MIT	Strong instruction-following; transparent benchmark suite	Weak quantitative reasoning; short context
FinTral	Mistral-7B	Continued pretraining + FinSet benchmark (25+ datasets) (Bhatia et al., 2024)	<i>No explicit licence</i>	Parameter-efficient; strong domain coverage	Governance risk due to missing licence/documentation
FinLLaMA	LLaMA-based	52B tokens + 573K finance instructions (Xie et al., 2024)	Meta Community Licence	Broad dataset/task coverage; strong in SA/NU	Licence restrictions (AUP); hallucination risk
Llama-2	Meta LLaMA-2	General RLHF instruction-tuning (Touvron et al., 2023a)	Meta Community Licence	Widely adopted; strong general reasoning	No finance-specific training; weak numerics
Mistral	Mistral-7B	General instruction-tuning; no finance corpus (Jiang et al., 2023).	Apache-2.0	Fast inference; permissive commercial licence	Lacks finance optimisation; weak regulatory/numeric reasoning

Table 1: Comparison of Open-Source Finance LLM 's

2.3 Benchmarks and Metrics

This study evaluates the Finance LLM models on three criteria for MCDA. The first is **Performance**, which captures the growing but uneven set of public benchmarks. The second is **Operational Capability**, which is captured through the context window. The attribute determines whether long inputs such as earnings-call transcripts, SEC filings, or tables can be processed without truncation, which directly impacts summarisation accuracy and numeric reasoning reliability. The third is **Licence & Regulation**, frequently overlooked by academic papers but critical for banking. This evidence is usually assessed through licence terms and documentation.

2.3.1 Task Performance

Performance is categorised as a criterion in this study to quantify financial capabilities such as sentiment analysis, named-entity recognition, numeric understanding, text summarisation, stock-market prediction, and credit scoring, which are consistently reported across FLARE (FinMA), FinSet (FinTral), and Open-FinLLMs (FinLLaMA) suites. **Sentiment Analysis (SA)** is evaluated using Financial PhraseBank (FPB) and FIQA-2018, which offer short, annotated finance extracts from news and forums (Malo et al., 2014; Ma et al., 2018). The **Named-Entity Recognition (NER)** system requires domain-specific tag sets, including FinNER and FinSet (Bhatia et al., 2024). The **Numeric Understanding (NU)** evaluation system uses FinQA to assess correct arithmetic chain responses and subsequent conversational follow-ons through ConvFinQA (Chen et al., 2021; Chen et al., 2022). **Text Summarisation (TS)** is important for generating executive digests. TS is evaluated on two datasets, ECTSum, which queries models to condense lengthy earnings-call transcripts to bullet points, and EDTSum, which targets short editorial summaries of financial press releases and articles (Mukherjee et al., 2022; Xie et al., 2023). Two predictive pillars used in the task performance criterion are **Stock Movement Prediction (SMP)** and **Credit-Scoring (CS)**. SMP involves predicting short-term market direction from headlines, while CS assigns risk ratings to borrowers using standard credit datasets (Xie et al., 2023; Xie et al., 2024).

The inconsistent results from individual datasets led researchers to merge them into larger composite suites. FLARE (FinMA) unites the six NLP tasks with predictive tasks to achieve FinMA's wide scope although it provides minimal depth (Xie et al., 2023). FinSet (FinTral) collects more than twenty-five datasets across eight tasks (Bhatia et al., 2024). The Open-FinLLMs (FinLLaMA) includes fourteen task types and thirty datasets (Xie et al., 2024). The

study selects six pillars from the mentioned benchmarks, which include SA, NER, NU, TS, SMP, CS, because these tasks have public scores that match model performance across the five alternatives. The decision matrix remains balanced because headline generation and risk-explanation tasks appear in research papers but do not have complete coverage.

2.3.2 Operational Capability

A model's **Context Window** defines the maximum number of tokens it can handle before it starts truncating the input. The operational attribute stands as a critical factor because it affects the processing of long earnings call transcripts during text summarisation (e.g., ECTSum) and numeric reasoning that depends on multiple tables. The official model cards and repositories serve as the source for each model's advertised context length to maintain consistency between different alternatives (Touvron et al., 2023a; Jiang et al., 2023; Xie et al., 2024). The practical advantages of context expansion become less significant after reaching typical segment sizes, so the benefits are most notable when moving from 4k to 16k tokens, but additional increases in context length yield minimal improvements.

2.3.3 Licence & Regulation

The value of a strong benchmark performance becomes limited when licensing terms restrict commercial use or when the model card fails to include vital information. The assessment of five attributes for each model followed SR 11-7 and the upcoming EU AI Act: **Commercial Use, Derivative and Redistribution rights, Licence Type, Usage Restrictions, and Documentation Clarity**. The FinMA (MIT) and Mistral (Apache-2.0) licences allow users to redistribute and use models for commercial purposes (Xie et al., 2023; Jiang et al., 2023). The LLaMA family models, including Llama-2 and FinLLaMA, operate under Meta's Community Licence, which allows commercial use but imposes Acceptable Use Policies and conditional restrictions that need thorough compliance checks before adoption (Touvron et al., 2023a; Xie et al., 2024). The governance risk of FinTral exists because the weights are available, yet there is no explicit licence or full documentation, which creates uncertainty about lawful or safe deployment regardless of its task performance (Bhatia et al., 2024).

2.3.4 Reliability and known failure modes

Three reliability issues consistently appear in finance-specific LLMs. According to FinQA and ConvFinQA benchmarks, the models without external tool support demonstrate poor performance in multi-step arithmetic and table lookups. This results in incorrect calculations with high confidence outputs (Chen et al., 2021; Chen et al., 2022). According to broader

studies on generative model hallucinations, the FinSet benchmark from the FinTral study indicates that definition-style QA and financial text summarisation still contain hallucinations, which are plausible but unverified statements (Bhatia et al., 2024; Ji et al., 2023). The models demonstrate poor performance when processing unstructured or semi-structured inputs that include tables and figures. The performance of models on numerically intensive tasks continues to fall short of expectations even after pre-training and instruction-tuning (Xie et al., 2024).

The operational risk of truncation occurs when inputs surpass the maximum context window because the system silently cuts content, which damages factual accuracy in summarisation tasks and breaks arithmetic sequences in numerical reasoning. The risk requires treating the context window as an explicit operational attribute instead of embedding it within summarisation or numeric reasoning scores (Touvron et al., 2023a; Jiang et al., 2023; Xie et al., 2024).

3. METHODOLOGY

3.1 Research Design

The dissertation employs Multi-Criteria Decision Analysis (MCDA) framework to structure and address the decision problem. MCDA is particularly appropriate to this study as it allows both qualitative and quantitative judgments to be considered together (Belton & Stewart, 2002). Unlike other approaches, MCDA enables diverse criteria to be compared on a common basis without oversimplification. The process is organised into three stages: **Problem Formulation, Evaluation, and Appraisal.**

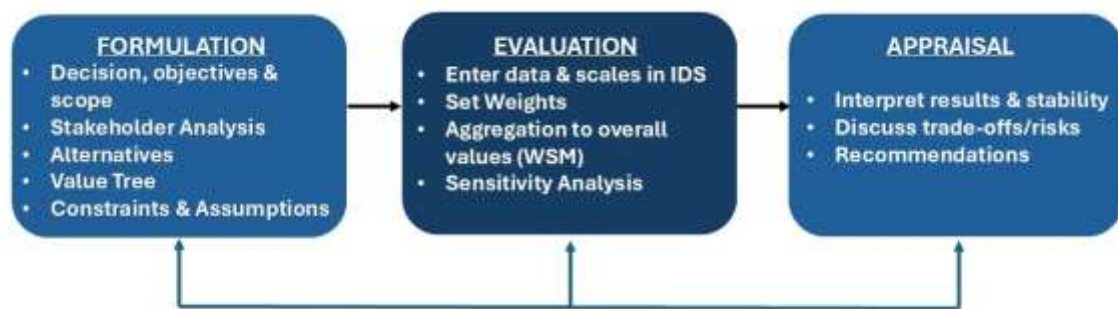


Figure 1: Stages of MCDA Framework

The study employs the literature review to identify the essential dimensions that determine the evaluation of finance-specific LLMs. The value tree presents the decision problem through three main branches, which include Performance, Operational Capability and Licence & Regulation. The finance industry priorities are reflected through measurable sub-criteria, which form the branches of the value tree. The study implemented non-compensatory constraints to maintain feasibility, which include requirements for open-source availability, documentation, reproducibility, and English language coverage. The evaluation process continued with models that satisfied these baseline requirements. The following chapter's analysis relies on alternatives that are both academically relevant and transparent and viable for real-world deployment.

3.2 Evaluation Method: Multi-Attribute Value Problem (MAVP)

This study uses an additive Multi-Attribute Value Problem (MAVP): each alternative “a” is assigned a single value obtained by summing its attribute values, each multiplied by its weight.

$$v(a) = \sum_{i=1}^m w_i v_i(a)$$

where $v_i(a) \in [0,1]$ is the value on criterion i and $w_i \geq 0$ are important weights with $\sum_i w_i = 1$ (Keeney & Raiffa, 1993).

To ensure compatibility across dimensions, the scores were rescaled to the $[0,1]$ interval using simple linear monotonic transformations (Hwang & Yoon, 1981). The qualitative attributes received ordinal values through a scoring system that placed OSI-approved licences above restrictive or unspecified licences. The procedure enabled the aggregation of diverse evidence types, including numerical benchmark scores and categorical governance conditions, into a unified decision-analytic framework (Keeney & Raiffa, 1993; Belton & Stewart, 2002).

The weights were derived by combining direct rating and swing weighting before being entered into Interactive Decision Support (IDS) software, which automatically normalised them to sum to one. Raw data were entered into IDS, rescaled through linear value functions, and aggregated using the Weighted Sum Model. Sensitivity analysis was then conducted in IDS to examine the robustness of rankings, using both one-way variation of weights and scenario-based weight allocations to reflect different stakeholder perspectives.

The additive MAVP method depends on several established assumptions for its validity. The first assumption states that attributes are **preferentially independent** because preferences for one criterion do not change based on the levels of another criterion (Keeney, 1992). The second assumption states that attributes demonstrate **monotonic behaviour** because their values either rise or fall steadily when their underlying criteria change. The third requirement for **commensurability** involved converting all attributes to a common 0–1 measurement scale through linear rescaling of continuous measures and min–max normalisation of the context window (Hwang & Yoon, 1981). The assumptions follow standard decision analysis procedures and create a solid theoretical framework for using the additive value model (Keeney & Raiffa, 1976; Belton & Stewart, 2002; Papamichail & French, 2013).

4. PROBLEM FORMULATION

4.1 Decision Statement and Objectives

This study is from a financial practitioner's perspective to choose one open-source large language model (LLM) for 2024–25 deployment, which provides maximum value in actual operational workflows. The selected model needs to show performance in finance-related tasks and operational reliability when processing intricate numeric and tabular documents.

The study uses multi-criteria decision analysis (MCDA) framework (Belton & Stewart, 2002; Hwang & Yoon, 1981) to prevent over-reliance on a single metric and to make trade-offs transparent. Licence & Regulation considerations are treated as eligibility filters, reflecting industry practice, rather than as the primary focus of regulatory compliance (Board of Governors of the Federal Reserve System, 2011; European Union, 2024).

The aim is to demonstrate a clear and defensible recommendation of the most suitable open-source LLM for finance workflows. The scope is limited to models that can be assessed using publicly available evidence such as academic studies, model cards, licences, and domain-specific benchmarks. Candidate models are evaluated through normalised and aggregated scores within an additive value model, with sensitivity analysis applied to test the robustness of results (Keeney, 1992; Hwang & Yoon, 1981).

4.2 Stakeholders' Perspective

The power–interest grid (Mendelow, 1991) is used to evaluate stakeholder influence on the model-selection process and to further interpret it through the power–legitimacy–urgency framework (Mitchell, Agle, & Wood, 1997). The visual representation of stakeholder placement appears in Figure 2, while Table 2 explains the reasoning behind each group and their corresponding engagement responsibilities.

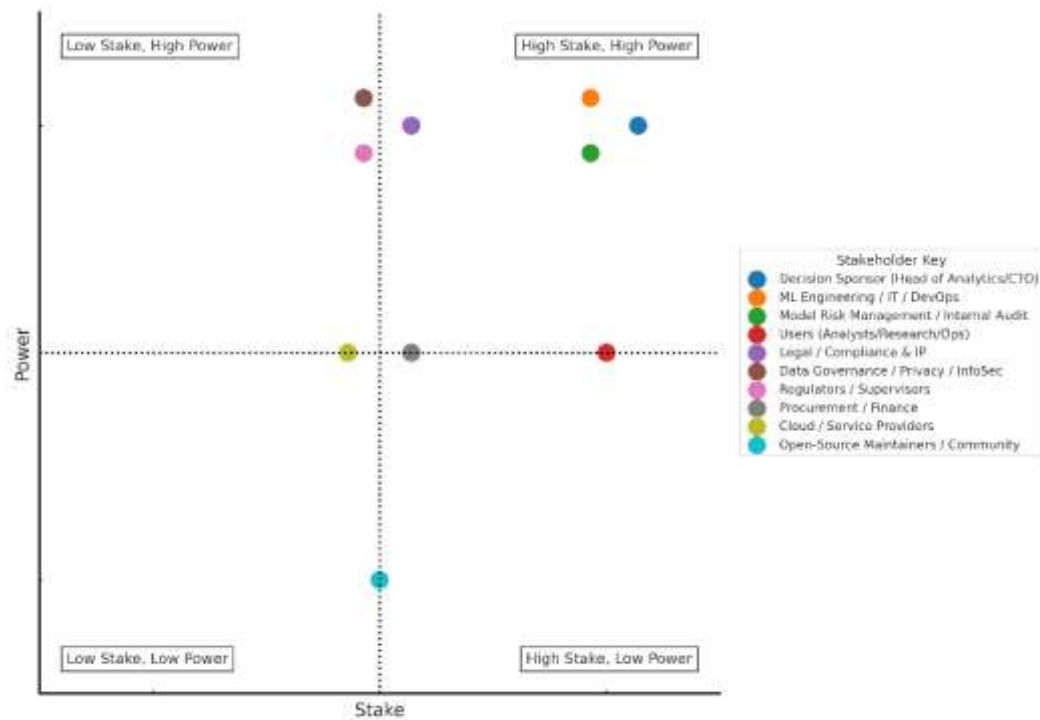


Figure 2: Stakeholder Analysis: Power vs. Stake

STAKEHOLDER	STAKE	POWER	EXPLANATION
Decision Sponsor (Head of Analytics/CTO/Product Owner)	High	High	Owns the business outcome and budget; authorises model selection and deployment.
Model Users (Analysts/Research/Operations)	High	Medium	Depend on model outputs for daily workflows; influence acceptance through usability and accuracy needs.
Model Risk Management (MRM) / Internal Audit	High	High	Gatekeeper for model approval, validation evidence, and documentation standards.
Legal/Compliance & IP	Medium	High	Assesses licence terms, redistribution/derivatives, and usage restrictions; can block non-compliant options.
Data Governance/Privacy & InfoSec	Medium	High	Sets data-access, logging, and privacy controls; ensures safe handling of PII and sensitive data.
ML Engineering / IT / DevOps	High	High	Own deployability, latency/throughput, and cost; critical for long-context (context window) operation.
Procurement / Finance	Medium	Medium	Manages contracting and total cost of ownership; influences timing and vendor terms.
Business Unit Clients / End-Users	High	Medium	Consumers of downstream insights; adoption depends on perceived reliability and usefulness.
Open-Source Maintainers / Community	Medium	Low	Provide updates, fixes, and documentation; limited decision power but affect sustainability.
Cloud / Service Providers	Medium	Medium	Constrain feasible infrastructure and pricing; impact performance for long-context workloads.
Regulators / Supervisors	Medium	High	Set boundary conditions via guidance (e.g., model-risk expectations); not direct selectors but can halt deployment.

Table 2: Roles and Influence of Stakeholders in Financial LLM Evaluation

In summary, the stakeholder analysis for finance-specific LLM adoption aims to establish a balance between technical requirements, regulatory needs, and operational considerations. The model approval and deployment process under established model-risk governance (Board of Governors of the Federal Reserve System, 2011) relies on high-power and high-stakeholders, including decision sponsors, Model Risk Management and engineering teams. The business units, analysts and end-users hold significant authority on adoption through their impact on workflow fit and reliability, which aligns with stakeholder salience theory (Mitchell, Agle & Wood, 1997). Legal, Compliance, and Data Governance functions maintain high power levels when licensing or privacy constraints become applicable (European Union, 2024; NIST, 2023), and regulators and supervisors establish boundary conditions that follow SR 11-7 and emerging AI governance norms. Open-source maintainers and infrastructure providers have lower power to influence the sustainability and feasibility aspects but do not determine fundamental choices according to Freeman’s (1984) expanded understanding of stakeholder interdependence. A RACI approach guides engagement to maintain balanced perspectives between stakeholders by making sponsors “*Accountable*” while analysts and engineers are “*Responsible*” and MRM and Legal are “*Consulted*” and governance bodies remain “*Informed*” (PMI, 2021). The sensitivity analysis uses performance-led and compliance-led weighting scenarios to demonstrate how stakeholder influence results in varied ranking outcomes (Belton & Stewart, 2002).

4.3 Alternatives

The process of defining alternatives serves as an essential requirement for MCDA as it establishes the set of feasible options that will be evaluated. The research defines five advanced large language models (LLMs) as potential alternatives. The alternatives differ from each other in their financial adaptation methods; however, they all meet the essential criteria of open accessibility, documented performance in financial or general NLP tasks, and institutional deployment readiness. The literature review demonstrates that these models represent the most suitable open-source candidates that balance domain specialisation, instruction alignment, licensing, and operational feasibility.

The first alternative is FinMA, a fine-tuned model derived from LLaMA with explicit alignment to financial tasks and released alongside the PIXIU benchmark (Xie et al., 2023). Mistral is included as a widely adopted instruction-tuned baseline released by Mistral AI (2023), serving as a competitive reference point in financial adaptation studies. To represent a larger-parameter general-purpose baseline, Llama-2 was selected, enabling assessment of

trade-offs between model scale and domain specialisation (Touvron et al., 2023a). In contrast, FinLLaMA provides a domain-tuned, instruction-aligned variant of LLaMA, specifically optimised for financial applications (Xie et al., 2024). Finally, FinTral represents a financial instruction-tuned model with an emphasis on regulatory and compliance contexts, making it particularly relevant for institutional use cases (Bhatia et al., 2024).

The research excluded closed-source systems including BloombergGPT and InvestLM. The training data and weights of BloombergGPT remain inaccessible to users which hinders transparent replication (Wu et al., 2023), and InvestLM focuses only on portfolio reasoning, which restricts its ability to compare with other models (Yang, Tang & Tam, 2023). The study excludes FinGPT as it operates as a learning framework instead of a benchmark model that can be used for MCDA (Yang et al., 2023).

The five selected alternatives include both domain-specialised financial LLMs and general-purpose baselines to ensure that the evaluation reflects the real-world trade-offs the finance industry encounters across task performance, governance, and operational feasibility.

4.4 Value Tree

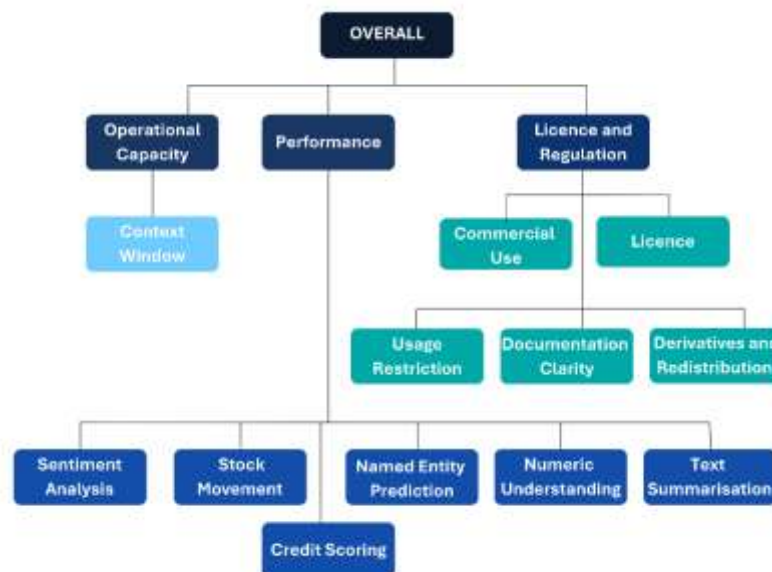


Figure 3: Value Tree

The value tree transforms the entire decision problem into a structured hierarchy of objectives together with criteria. The value tree illustrates how the general financial LLM selection goal can be divided into quantifiable components, which enhances evaluation transparency and systematisation. The value tree structure consists of three primary branches: Performance, Operational Capability, and Licence and Regulation.

The **Performance** branch evaluates the technical capabilities of models when performing essential financial operations. The evaluation framework consists of six sub-criteria that focus on financial institution applications, including Sentiment Analysis (SA), Named Entity Recognition (NER), Numeric Understanding (NU), Text Summarisation (TS), Stock Movement Prediction (SMP), and Credit Scoring (CS). The combination of these tasks demonstrates both analytical reasoning and predictive functions, which offer a complete assessment of model capability.

The **Operational Capability** branch evaluates the practical deployment potential of models. The study focuses on the context window as it establishes the maximum amount of input data that a model can process simultaneously. The finance industry requires this criterion as decisions in this field depend on extensive documents, structured data, and complex reasoning processes (Srivastava et al., 2024). The inclusion of this criterion was necessary because analysts, auditors, and compliance teams work with extensive disclosures, yet limited context windows create obstacles for practical deployment.

The **Licence & Regulation** branch implements governance principles that determine the permissible scope of model deployment. The sub-criteria of this branch include Commercial Use, Derivatives and Redistribution, Licence Type, Usage Restrictions, and Documentation Clarity. The sub-criteria represent the legal and compliance requirements that institutions must follow during their operations. The criteria match the requirements of SR 11-7 (Board of Governors of the Federal Reserve System, 2011) and the EU AI Act (European Commission, 2024). The differences between these licences create substantial financial impacts. Permissive licences enable seamless lawful deployment and downstream reuse, but restrictive or absent licences elevate governance risk and compliance expenses (Board of Governors of the Federal Reserve System, 2011; European Union, 2024).

4.5 Key Values and Attributes

Value	Attribute	Importance
Performance	Sentiment Analysis	Evaluates how effectively a model detects tone and polarity in financial texts such as news or analysts' research reports. This attribute is essential for risk assessment and market sentiment tracking.
	Name Entity Recognition	Evaluates whether the model can accurately identify financial entities such as companies, instruments, regulatory bodies, currencies, etc. This attribute is essential for structured data extraction and compliance reporting.
	Numeric Understanding	Evaluates whether the model can accurately interpret numbers embedded in financial texts such as ratios, growth rates, etc. This attribute is essential for ratio analysis, earnings interpretation and numerical reasoning.
	Text Summarisation	Evaluate whether a model can summarise long financial disclosures into concise, accurate summaries. This attribute is essential for decision support, analyst screening, and risk monitoring.
	Stock Movement Prediction	Evaluate whether a model can link unstructured financial text with price movements. This attribute is essential for facilitating trading decisions, forecasting and risk management.
	Credit Scoring	Evaluate whether a model can translate financial information into relevant risk assessments. This attribute is essential for lending decisions, compliance checks, and regulatory reporting.

Operational Capability	Context Window	Evaluates how much information a model can process in one sequence. This attribute is essential for complex reasoning and comprehensive financial reports.
Licence and Regulation	Commercial Use	Evaluates whether the model can be implemented in real-world financial departments/institutions. This attribute is essential for adoption of model, operational reliability and deployment within the business's ecosystem.
	Derivatives and Redistribution	Evaluates whether the models or their fine-tuned versions can be shared. This attribute is essential for knowledge transfer and future innovations.
	Licence	Evaluate the legal obligations and restrictions associated with this model. This attribute is essential for compliance, adoption feasibility and continued relevance.
	Usage Restrictions	Evaluates whether the model has any specified restrictions such as prohibitions on financial services. This attribute is essential as restrictions can cause models to be disqualified from the decision-making process.
	Document Clarity	Evaluate the transparency of the information disclosed about the model. This attribute is essential for auditability and maintaining compliance requirements.

Table 3: Key Values and Attributes

4.6 Constraints

To ensure that the study is relevant, it is important to establish the constraints that the models need to satisfy. These constraints function as non-compensatory conditions, which means that if an alternative fails to satisfy the hard constraints, it cannot be considered for further analysis, regardless of its strengths in other criteria or areas (Belton & Stewart, 2002). The following three constraints represent minimum requirements for feasibility, transparency, and relevance in a financial environment.

The first is the **Open-source availability** of the models. Since closed source models restrict access, decrease transparency, and restrict vendor flexibility, the models that are not released under open or permissive licences are not included in this study (Wu et al., 2023; Xie et al., 2024).

The second is **Documentation and Reproducibility**. To ensure that the performance of models can be independently verified, the models must provide model cards, technical documentation, and benchmarks. If this constraint is not achieved, the institutions would not be able to satisfy regulatory requirements for auditability and transparency (Board of Governors of the Federal Reserve System, 2011; Kang & Liu, 2023).

The third constraint is **English language coverage**. As most financial disclosures, regulatory filings, and market data are predominantly produced in English, this study only considers models that satisfy this constraint.

These constraints combined guarantee that the alternatives chosen in this study are viable, transparent, and relevant to the finance industry.

5. EVALUATION

5.1 Data Sources and Method

The research utilises financial NLP benchmarks, operational capabilities, and licence documentation to obtain evaluation data. These sources offer a systematic framework to evaluate advanced language models in finance.

The Performance criteria scores were obtained from a combination of benchmark tests and evaluation suites. FinLLaMA is evaluated by the FinBEN benchmark, which consists of six tasks and fifteen datasets. The FinBEN benchmark provides extensive testing of sentiment analysis, entity recognition, numerical reasoning, and classification tasks (Xie et al., 2024). FinMA uses scores from the FLARE benchmark, which assesses the model across multiple skills and capabilities (Xie et al., 2023). The FinSet benchmark serves as the evaluation for FinTral as it represents a domain-specific assessment tool published with the models' release (Bhatia et al., 2024). To ensure reliability, the scores were also evaluated against established benchmarks such as FIQA-SA for sentiment analysis (Maia et al., 2018) and the Financial PhraseBank for entity recognition (Malo et al., 2014). The selected benchmarks focus on financial domain tasks as they help demonstrate performance on specialised financial applications instead of general NLP. This distinction is important in finance as the models need domain-specific languages combined with numerical reasoning and structured reporting, which general-purpose benchmarks fail to capture.

The operational capability was represented by a context window, which reflects the ability of the model to analyse financial disclosures, long reports, and multi-document inputs. Model cards, Hugging Face, and GitHub repositories provided the scores for this criterion (Mistral AI, 2023; Meta AI, 2023). In finance, this evaluation is relevant as decision-making often depends on processing long and complex documents, and a limited context window may restrict the model's real-world deployment.

The licence & regulation criteria received scores from licence agreements, model documents, and official repositories. FinMA is released under the permissive MIT licence, whereas Mistral is released under Apache-2.0. Both these licences are OSI-approved and allow broad commercial use and redistribution. In contrast, Llama-2 is released under a community licence that allows the models to be used commercially while placing limits on redistribution and implementing Acceptable Use Policies (Meta AI, 2023). FinTral does not release a licence for its fine-tuned weights, so the model outputs are effectively restricted to research purposes and

lack formal usage rights (Bhatia et al., 2024). For finance, the contrast of model licence is critical as open licences ease compliance and support reuse, whereas a restrictive or absent licence introduces challenges and may limit adoption under regulatory standards (Board of Governors of the Federal Reserve System, 2011; European Union, 2024).

5.2 Weight Elicitation

IDS provides a visual weighting tool that enables users to determine weights for the established criteria. The standard MCDA practice required interactive weight adjustments until the criteria matched the specific decision environment. Through this approach, the decision makers could translate their subjective judgments into numerical values while maintaining structural consistency (Keeney & Raiffa, 1993; Belton & Stewart, 2002).

The performance criterion received the greatest weight allocation at 0.65, as decision makers, including the Head of Analytics, CTO, analysts, and business units, consider financial accuracy their main source of value. The Legal and Compliance team maintains governance control through SR 11-7 and the EU AI Act and hence receives 0.20 as their highest weight allocation (Board of Governors of the Federal Reserve System, 2011; European Union, 2024). Engineers and IT Teams support operational capability with 0.15 weight as they focus on both feasibility and throughput, although large context windows provide minimal extra value beyond a threshold (Mendelow, 1991; Mitchell, Agle & Wood, 1997; Freeman, 1984).

Assessing the sub-criteria within the performance branch, text summarisation and credit scoring received greater weight allocations due to their impact on analyst efficiency and auditability, whereas sentiment analysis, entity recognition, numeric understanding, and stock movement prediction were given equal weight. The operational capability branch contained only the context window, which received its assigned weight. Within the licence and regulation branch, commercial use and licence type were prioritized in accordance with Legal and Compliance concerns, whereas redistribution rights, usage restrictions, and documentation clarity received gradually diminishing shares.

IDS automatically normalised the weights to maintain consistency. The numerical weights linked to stakeholders' priorities and regulatory expectations allowed the process to combine analytical rigor with the practical realities of model deployment in the finance industry.

5.3 Empirical Results

Through IDS, the scores of all five alternatives across the three branches were aggregated. The results indicated a distinct demarcation between the finance-specialised and general-purpose models with differences in performance, licence clarity, and operational feasibility that affect the final ranking.

5.3.1 Overall Ranking

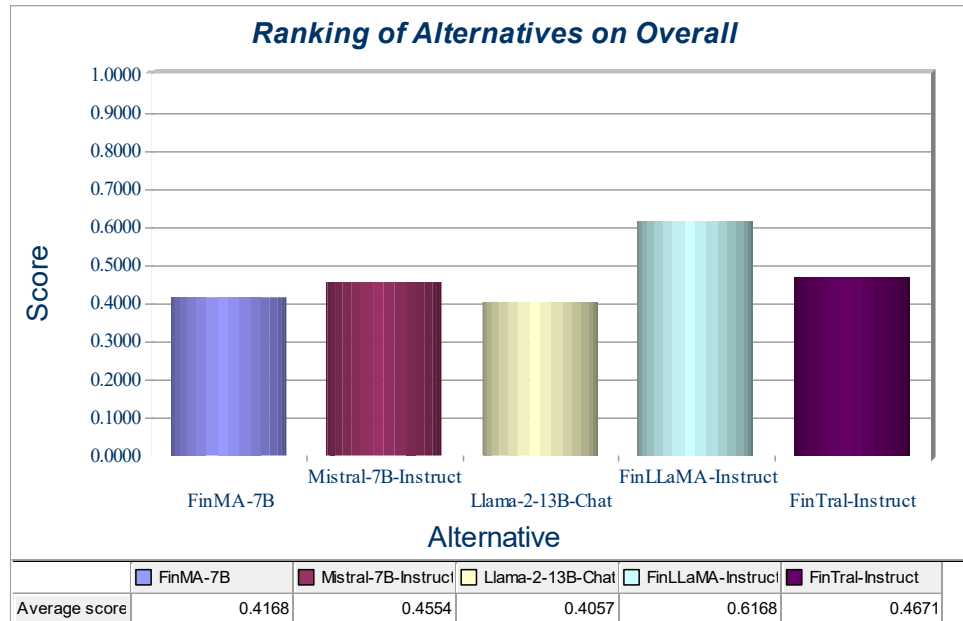


Figure 4: Overall Ranking

As illustrated in Figure 4, the ordering clearly indicates that the ranking favours finance-specialised models. FinLLaMA is top-ranked due to its consistent score across benchmarks, while FinTral's task-focused fine-tuning resulted in better performance of specific tasks. In comparison, Mistral and FinMA advanced from permissive licences, but their scores on the finance task performance were restrictive. Llama-2 performed poorly on both criteria and was surpassed by other models.

5.3.2 Performance

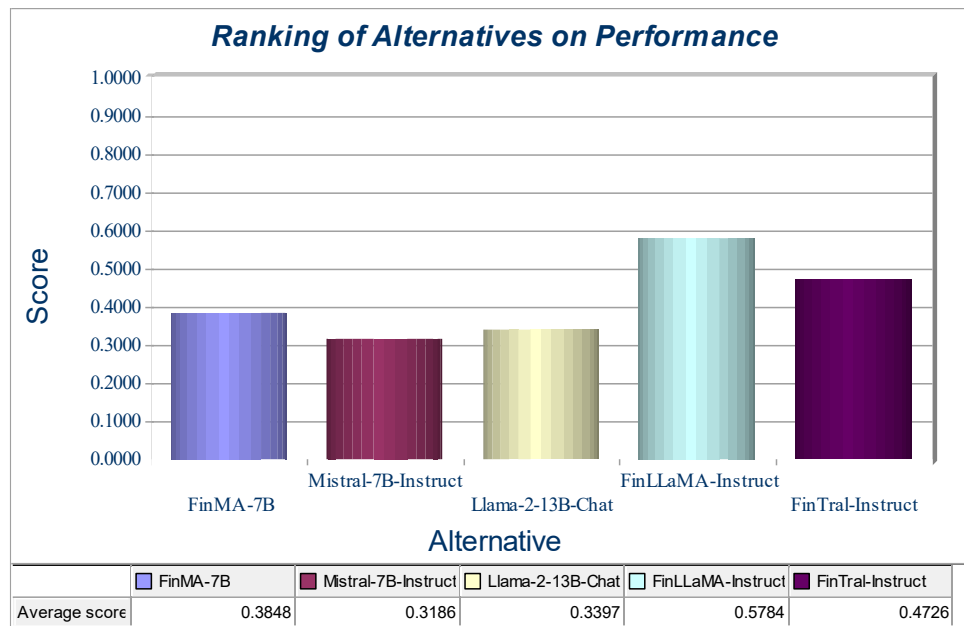


Figure 5: Performance Ranking

The performance scores are illustrated in Figure 5. FinLLaMA ranked the highest with 0.578, followed by FinTral at 0.473. At the sub-criterion level, distinct patterns were observed. (i) FinMA ranked highly in sentiment analysis (0.88) while underperforming in other tasks. (ii) FinLLaMA scored the highest in numeric understanding, while maintaining a balanced score across other tasks. (iii) FinTral performed the best in summarisation (0.40) and credit scoring (0.61) which reflected in its fine-tuning for risk assessment workflows. Mistral and Llama-2, in comparison to other alternatives, underperformed in entity recognition and numeric reasoning which limited their value for compliance-related tasks.

5.3.3 Operational Capability

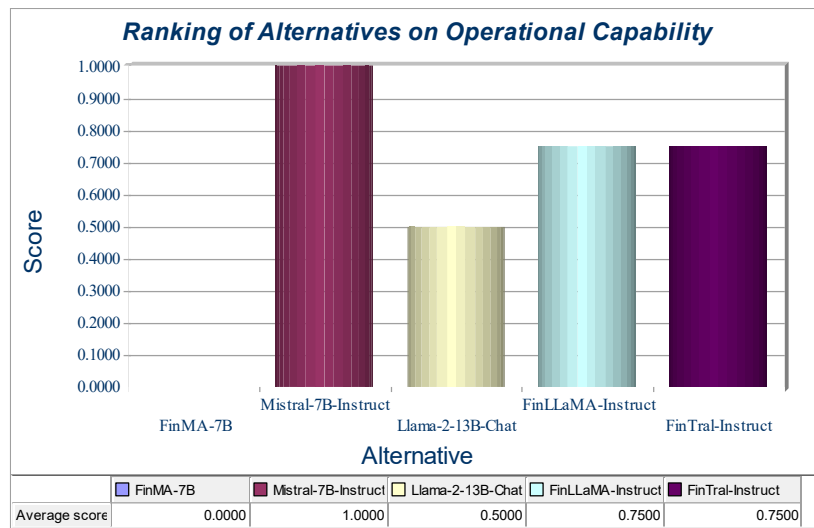


Figure 6: Operational Capability Ranking

The operational capability scores, measured by context window, are shown in Figure 6. Mistral-7B ranked first due to its extended context capacity, whereas FinLLaMA and FinTral followed next at 0.75 each. The relatively low weight for this branch meant that operational capability was insufficient to counter the weaknesses of the alternatives in benchmark performance or licensing.

5.3.4 Licence & Regulation

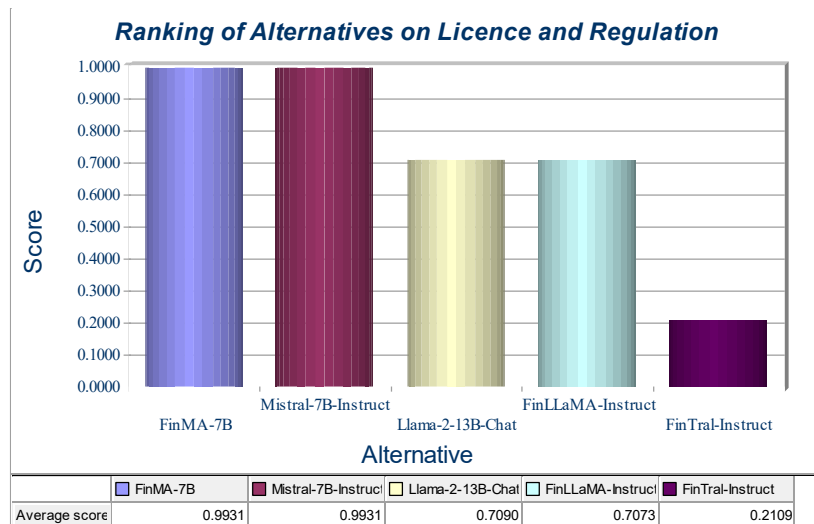


Figure 7: Licence and Regulation Ranking

As indicated in Figure 7, the scores in Licence & Regulation reorder the ranking. Due to permissive MIT and Apache-2.0 licences, FinMA and Mistral achieved close to perfect scores of 0.993. FinLLaMA (0.707) and LLaMA-2 (0.709) were constrained due to Meta's community

licences that imposed redistribution limits and Acceptable Use Policies. FinTral ranked the lowest due to the absence of a published licence, which makes its deployment in the finance industry infeasible despite its strong performance in benchmark tasks. The Licence & Regulation scores show that licensing clarity is crucial for compliance.

5.4 Sensitivity Analysis

In this study, sensitivity analysis is performed to examine the robustness of the decision-making process and to assess the impacts on ranking due to changes in weights. In accordance with Belton and Stewart (2002), three approaches are applied: (i) one-way sensitivity analysis, (ii) threshold analysis, (iii) scenario analysis. Threshold analysis is integrated with one-way plots since the crossover of alternatives is distinctly visible in the same plots.

5.4.1 One-Way Sensitivity and Threshold Analysis

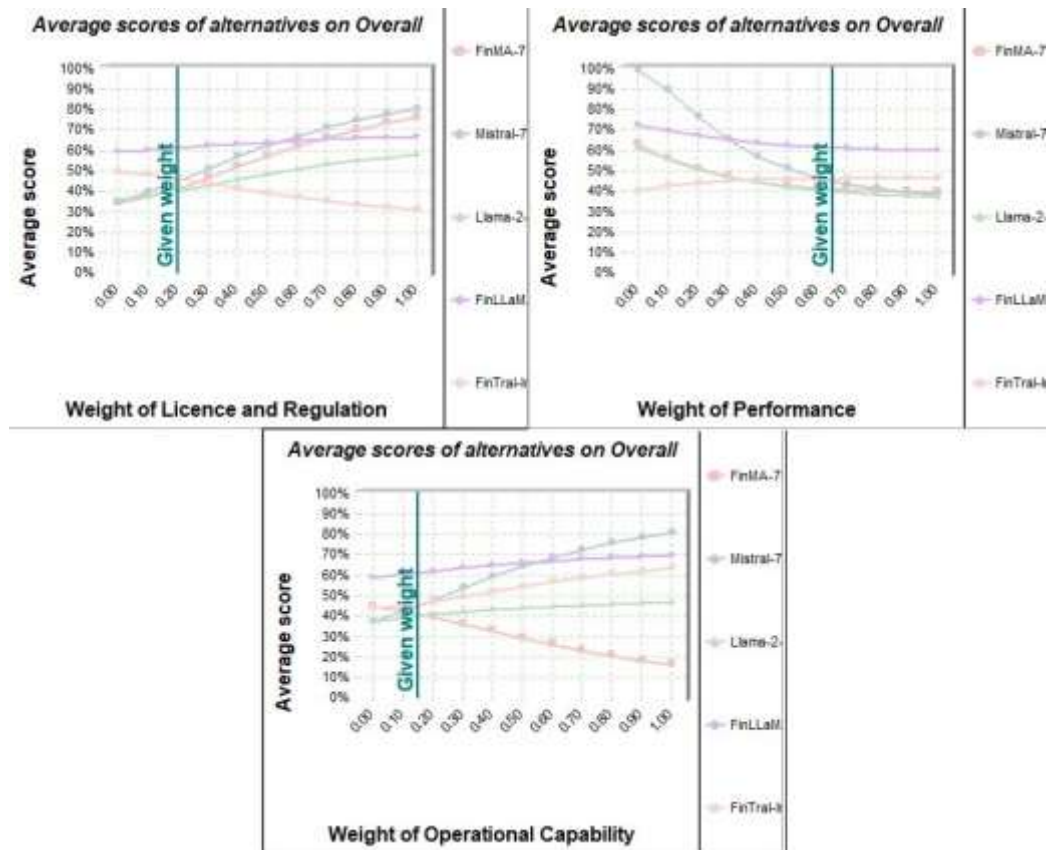


Figure 8: One-way sensitivity analysis across top-level criteria weights.

The weights of the top-level criteria were adjusted individually. FinLLaMA consistently retains the leading position, demonstrating the robustness of the decision-making process under the performance criterion. In contrast, a sensitive pattern was observed under the operational capability, when the weights exceeded 0.50, Mistral overtook FinLLaMA. This demonstrates its high score on this criterion. Similarly, as the weight of Licence & Regulation is increased

upwards of 0.45, FinLLaMA is surpassed by FinMA and Mistral, resulting from permissive licensing. The crossover points indicate the threshold effects, where changes in the weight acts as catalysts for reversal ranking. In conclusion, FinLLaMA is robust for performance-led assumptions; the ranking is vulnerable if licence or operational capability is given excessive importance.

5.4.2 Scenario Analysis

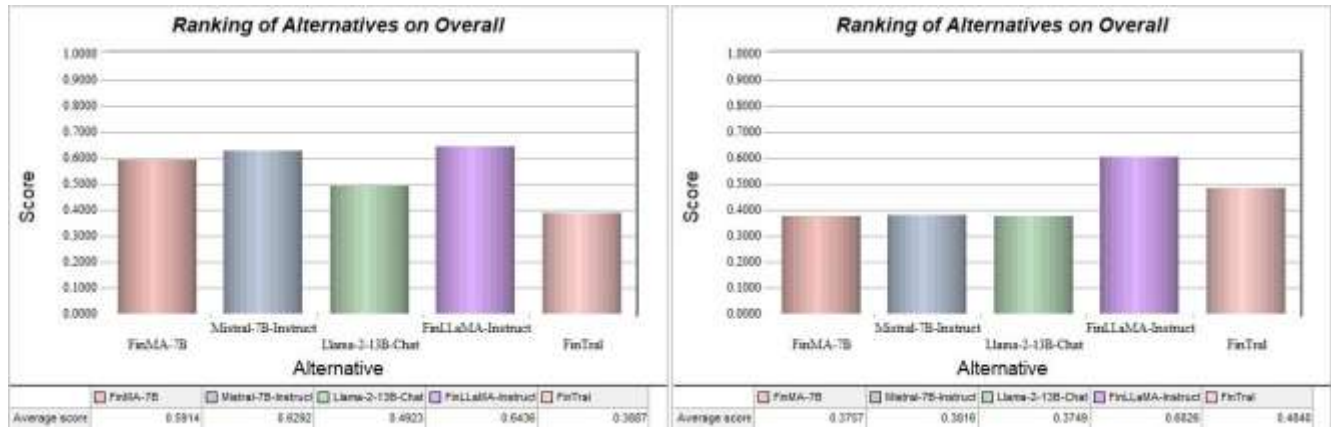


Figure 9: Comparison of model rankings in performance-led and compliance-led scenarios.

Scenario analysis is conducted to reflect stakeholders' perspectives. In a compliance-led scenario, the weights are allocated as 0.50 (Performance), 0.10 (Operational Capability), and 0.40 (Licence & Regulation). These weights were strategically allocated in line with the rationale of legal regulators and risk management, who consider Legal & Regulation as important as Performance. In this scenario, FinLLaMA (0.6436), Mistral (0.6292), and FinMA (0.5914) clustered towards the upper end, while FinTral severely underperformed, indicating that Licence permissiveness played an essential role in the outcome.

In a performance-led scenario, the weights are allocated as 0.75 (Performance), 0.15 (Operational Capability), and 0.10 (Licence & Regulation). These weights were strategically allocated in line with business units and analytics teams' rationale, prioritising accuracy over governance. In this scenario, FinLLaMA (0.6026) maintains the top rank, while FinTral (0.4840) is in the second place, which highlights the model's improvements from targeted fine-tuning. The remaining models significantly underperformed. These shifts demonstrate how different stakeholder perspectives impact the scores, increasing or decreasing the gap between the models. The robustness of the decision-making process is highlighted with FinLLaMA maintaining the leading position consistently.

6. APPRAISAL

6.1 Summary of Results

As seen in empirical results, there is a distinct separation between finance-specialised and general-purpose models. FinLLaMA steadily maintains its position as a top-ranking model, with a balanced performance criterion indicated by its sentiment analysis, numeric reasoning, and summarisation scores. FinTral emerged as the second-ranked alternative with substantial results in performance NLP tasks such as credit scoring and summarisation. However, the lack of a published licence restricted the model's suitability for a real-world deployment. Mistral and FinMA benefited from permissive licensing, however their underperformance in benchmark scores restricted their competitiveness in overall ranking. In contrast to the above-mentioned models, LLaMA is the weakest model across both performance and governance criteria.

The results highlight the trade-off in the model selection, where it can be noticed that strong benchmark performance alone is not sufficient if there is restrictive governance or licensing, and conversely, licensing cannot compensate for the benchmarks in performance criteria.

6.2 Synthesis of Sensitivity Analysis

FinLLaMA consistently ranked highly in performance-led weights but with a shift in weights for operational capability and licence & regulation criteria, Mistral and FinMA surpassed the model. The crossover points in the plots mentioned in the sensitivity analysis indicate that ranking reversals are solely dependent on stakeholders' rationale.

As seen in scenario analysis, FinLLaMA consistently remained in leading positions in both scenarios. In a compliance-led scenario, FinLLaMA, Mistral and FinMA clustered at the top, whereas FinTral fell sharply owing to a lack of a licence. In a performance-led scenario, FinLLaMA's position was validated, with FinTral's progressing to second position due to task fine-tuning. The other models performed significantly worse.

From the results, we can conclude that if licensing or operational capability are given disproportionate weights, the rankings will be unpredictable. Furthermore, stakeholders' rationale has a significant impact on the ranking.

6.3 Robustness of the results and limitations

The study results show reasonable stability as FinLLaMA maintains its strong position in most weighting scenarios. The reliability and breadth of findings face multiple limitations that affect the study. A major concern lies in the inconsistency of benchmarks across the finance-specific LLM literature. The research papers in this field create their own benchmark sets, which frequently connect to their training data and evaluation objectives. The assessment of competing models occurs through different finance tasks, hindering their direct comparison.

The evaluation parameters receive inconsistent coverage throughout the assessment process. The publications typically present results for sentiment analysis and summarisation but frequently omit essential metrics, including the hallucination index, the carbon footprint, and the headline underperformance. The analysis became restricted because many complete values remained unpublished. The decision model required researchers to combine scattered evidence through a lengthy process that omitted certain evaluation criteria.

6.4 Practical Implications

For the finance industry, these results highlight the need to align model adoption with strategic priorities. When accuracy on financial NLP tasks is the key concern, FinLLaMA is distinguished as the strongest and most defensible choice. In contrast, when regulatory compliance and licence clarity take precedence, permissively licenced models such as FinMA and Mistral may offer more practical solutions, even though their task performance is weaker. FinTral shows the advantages of targeted fine-tuning for domain-specific accuracy, but its lack of licence transparency makes it unsuitable under current governance requirements. Ultimately, the finance industry must strike a balance between technical performance and compliance assurance, recognising that governance constraints can sometimes outweigh performance gains.

6.5 Reflection Against Literature

The current study supports recent findings that highlight the trade-off between domain specialisation and deployment feasibility. Research by Kang and Liu (2023) and Xie et al. (2024) shows that models receiving fine-tuning achieve superior task performance, yet face challenges with deployment and regulatory compliance. Bhatia et al. (2024) demonstrate that permissive licensing strongly influences institutional adoption rates, which is consistent with this study's results, where FinMA and Mistral performed better in a compliance-led scenario. Similarly, Xie et al. (2023) presented FinMA within the PIXIU benchmark while emphasising

openness as a key differentiating factor, helping to explain its stronger performance under governance-heavy weightings. By contrast, the limitations of FinTral mirror broader issues facing finance-specific LLMs, particularly regarding transparency and reproducibility (Cheng et al., 2024).

The combined assessment shows that task performance stands as a vital differentiator, yet licensing clarity and operational feasibility determine the actual practicality of financial LLMs. The evaluation of models for financial services requires a balanced approach that combines technical and governance perspectives to support decision-making.

6.6 Conclusion

This study shows that performance, licence clarity, and operational feasibility each have distinct but sometimes competing effects, creating trade-offs in model choice. FinLLaMA was consistently top-ranked and most reliable option when accuracy on financial tasks is the main aim. In compliance-led scenarios, FinMA and Mistral move up the rankings because of their permissive licensing, even though their benchmark scores were weaker. FinTral demonstrated the value of domain-specific fine-tuning, but its lack of transparent licensing raised serious governance concerns.

Overall, the findings highlight that performance as a standalone factor is inadequate to ensure deployment readiness in the finance industry. Decision-making requires a balance of technical capability, governance obligations, and operational practicality. Using the MCDA framework helped make these trade-offs visible in a transparent way, giving a defensible basis for selecting the most appropriate financial LLM.

7. CONCLUSION

The study aimed to assess open-source large language models (LLMs) for financial applications through a structured multi-criteria decision analysis (MCDA) framework. The decision problem was framed from a financial practitioner's perspective, to select models which deliver operational value through accurate financial tasks while ensuring compliance and feasibility standards. The study employed a decision-analytic perspective to deliver academic knowledge and operational guidance about financial LLM implementation.

The research reveals that performance, licence & regulation, and operational feasibility act independently but sometimes against each other to determine model suitability. The results show that FinLLaMA stands out as the most dependable choice for financial accuracy needs. The models FinMA and Mistral gained prominence in compliance-led scenarios as a result of their permissive licensing, though they underperformed in performance benchmarks. FinTral showed the advantages of domain-specific fine-tuning, yet its vague licensing terms created a major governance challenge. The results demonstrate that benchmark performance does not guarantee deployment readiness as the financial industry decision-making needs to balance technical capabilities with governance requirements and operational feasibility.

The research proves that MCDA enhances the evaluation of financial AI through its ability to assess multiple models beyond individual ranking systems. The framework revealed hidden trade-offs while allowing stakeholders to test different perspectives and establishing a clear method for selecting models that are both transparent and defensible. The research combined academic evaluation methods with practitioner needs to demonstrate how structured decision tools help organizations make sense of complicated AI adoption choices.

The study presents three essential findings. Firstly, the study enhances academic knowledge through its approach to evaluate LLMs by treating them as decision-making problems instead of focusing solely on technical benchmarking. Secondly, the study delivers useful information to financial organizations by revealing both the advantages and management constraints of prominent open-source models. Thirdly, the research demonstrates how MCDA enables organizations to make decisions based on evidence when they need to balance multiple competing priorities.

Nonetheless, several limitations must be acknowledged. The evaluation process depended solely on publicly available information, which limited the ability to perform thorough assessments and maintain equivalent evaluation standards. The benchmarking results across

the finance-specific LLM literature are fragmented and incomplete in terms of important metrics such as hallucination risk, computational efficiency, and environmental footprint. The conclusions therefore reflect the best available but incomplete evidence base.

The field of AI financial model evaluation presents several research directions for future investigation. Standardized financial benchmarks need development to achieve model comparison consistency as current evaluation methods create evaluation fragmentation. The assessment criteria need to expand to include hallucination risk together with explainability, computational efficiency, environmental impact, and task performance. The practical application of research results depends on conducting pilot tests of financial institutions to validate their findings. Research studies that compare open-source models to proprietary systems at scale would help scientists understand how transparency and licensing affect performance.

In conclusion, the study demonstrates that financial LLM adoption requires financial organizations to weigh between performance excellence, governance requirements, and operational practicality. The MCDA framework provides both academic excellence and practical decision support for financial organizations to select AI solutions transparently in their evolving digital environment.

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9. APPENDIX

Appendix A -Selection and Characteristics of Finance LLMs

A.1 Comparative Overview of Open-Source and Baseline LLMs

Model	Base Architecture / Parameters	Release (Year)	Training Scope / Data	Context Window	Benchmark Coverage	Licence Type	Sources
FinMA-7B	LLaMA-7B backbone, fine-tuned	2023	128K finance-specific instructions across 6 NLP + 2 predictive tasks (credit scoring, stock movement)	~4k tokens (inherits LLaMA-7B limits)	FLARE benchmark (SA, NER, NU, TS, SMP, CS)	MIT (permissive, OSI-approved)	Xie et al. (2023)
FinTral-Instruct	Mistral-7B with domain-specific continued pretraining + DPO	2024	Trained on FinSet: 25+ datasets, 8–9 task categories	16k tokens (Mistral extension)	FinSet benchmark (domain-wide financial tasks)	No explicit licence (weights released without terms)	Bhatia et al. (2024)
FinLLaMA-Instruct	LLaMA backbone, instruction-aligned	2024	52B financial tokens, 573K finance-specific prompts (SEC filings, macro indicators, market data)	16k tokens	Open-FinLLMs benchmark (14 tasks, 30+ datasets)	Meta Community Licence (restrictive AUP, redistribution limits)	Xie et al. (2024)
Llama-2-13B-Chat	Meta LLaMA-2, 13B parameters, RLHF-tuned	2023	General instruction tuning across multi-domain conversational & reasoning tasks	4k / 32k (depending on release variant)	General NLP benchmarks; not finance-specific	Meta Community Licence (AUP)	Touvron et al. (2023a)
Mistral-7B-Instruct-v0.1	Mistral-7B dense transformer, 7B parameters	2023	General-purpose instruction tuning (no finance corpus)	8k–16k tokens (sliding window)	General NLP benchmarks	Apache-2.0 (permissive, OSI-approved)	Jiang et al. (2023)

Table A.1: Comparative Overview of Open-Source Finance oriented and Baseline LLMs

A.2 Excluded Models and Rationale for Omission

Model	Base Architecture / Parameters	Release (Year)	Open / Closed	Primary Focus / Training Scope	Reason for Exclusion	Sources
BloombergGPT	50B parameter Transformer, finance-specific	2023	Closed-source (weights and data not released)	Trained on a proprietary dataset of Bloomberg’s financial documents and news	Not open-source; weights and training data unavailable, preventing reproducibility or transparent evaluation	Wu et al. (2023)
FinGPT	Framework built on smaller LLM backbones (varies)	2023	Open-source (framework, not a standalone model)	Designed as a continual learning framework that adapts smaller models incrementally	Not a fixed benchmarked model; unsuitable for direct comparison in MCDA	Yang et al. (2023)
InvestLM	LLaMA-based, fine-tuned for finance	2023	Open-source (research release)	Optimised for portfolio reasoning and investment-specific tasks	Narrow scope; lacks coverage of broader NLP benchmarks (e.g., SA, NER, TS)	Yang, Tang & Tam (2023)

Table A.2: Excluded Proprietary and Limited-Scope Finance LLMs

A.3 Finance-Specific NLP Tasks and Benchmarks

Task	Dataset(s)	Benchmark Suite(s)	Description / Use
Sentiment Analysis (SA)	Financial PhraseBank (FPB), FIQA-SA	FLARE, FinSet, Open-FinLLMs	Annotated financial news and forum excerpts used to evaluate tone and polarity detection.
Named Entity Recognition (NER)	FinNER, FinSet-NER	FinSet, Open-FinLLMs	Domain-specific tagging of financial entities (companies, regulators, instruments, currencies).
Numeric Understanding (NU)	FinQA, ConvFinQA	FLARE, FinSet, Open-FinLLMs	Evaluates multi-step reasoning, arithmetic accuracy, and table-based calculations.
Text Summarisation (TS)	ECTSum, EDTSum	FLARE, FinSet, Open-FinLLMs	Long-form (earnings-call transcripts) and short-form (editorial disclosures) summarisation benchmarks.
Stock Movement Prediction (SMP)	Headline-based prediction datasets (e.g., FiQA headline tasks)	FLARE, Open-FinLLMs	Predicts short-term market direction from unstructured text inputs.
Credit Scoring (CS)	Standardised credit datasets (PIXIU benchmark, FinSet credit-scoring tasks)	FLARE, FinSet	Evaluates ability to assign borrower risk ratings from structured + textual inputs.

Table A.3: Finance-Specific NLP Tasks, Datasets, and Benchmark Suites

Appendix B- Weight Elicitation and MAVP Assumptions

B.1 Weight Elicitation

B.1.1 Top-Level Criteria Weights

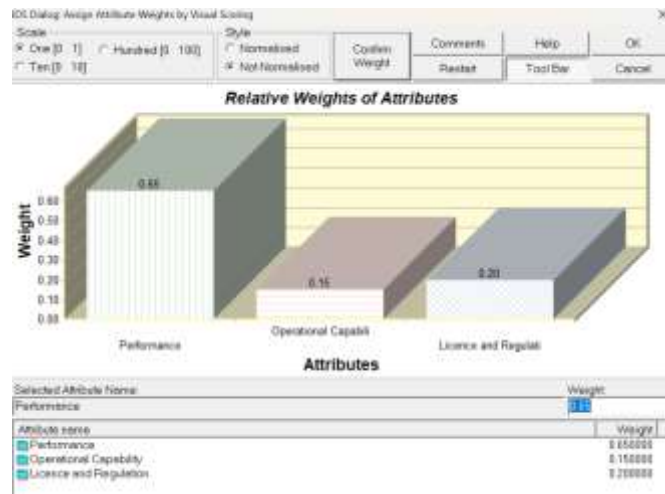


Figure B.1.1: Weights assigned to Top-Level Criteria

B.1.2 Performance

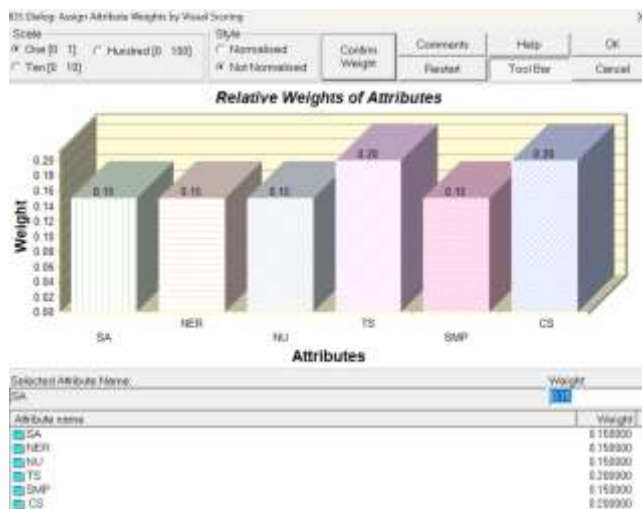


Figure B.1.2 Performance Sub-Criteria Weights

B.1.3 Operational Capability

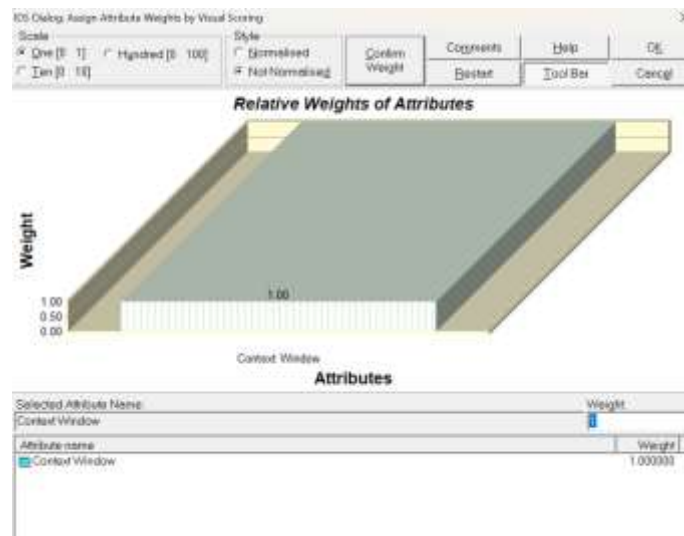


Figure B.1.3 Operational Capability Sub-Criteria Weights

B.1.4 Licence & Regulation

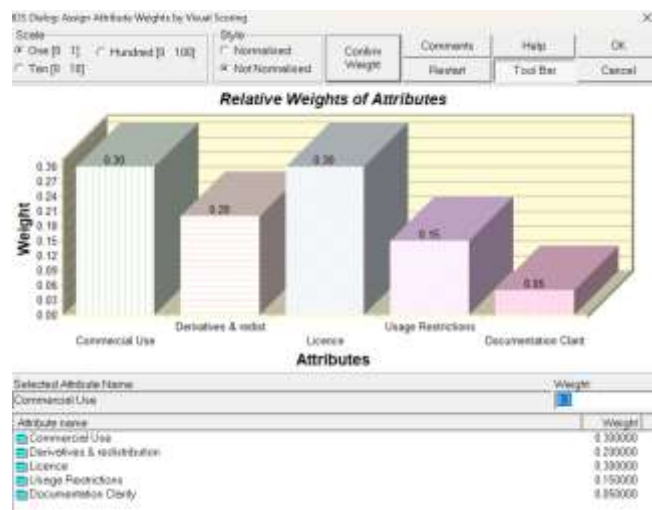


Figure B.1.4 Licence & Regulation Sub-Criteria Weights

B.2 MAVP Assumptions

The evaluation used a Multi-Attribute Value Problem (MAVP) framework with an additive value model that relied several fundamental assumptions for its operation. The evaluation team verified all assumptions to confirm that the chosen methodology remained suitable for assessing financial LLMs.

B.2.1 Preferential Independence

The concept of preferential independence states that preferences for one criterion remain unaffected by the values of other criteria. The stakeholders evaluated performance task importance through independent assessments of numeric reasoning and summarization tasks, regardless of the licence type. The stakeholders maintained a consistent distinction between technical accuracy and governance requirements which met this condition.

B.2.2 Monotonicity

The assumption of monotonicity states that any increase in desirable attributes leads to higher preference levels. The research study treated higher benchmark accuracy, longer context window capacity, and fewer licence restrictions as positive improvements. The criteria in this study did not present any inverse relationships between variables, so the monotonicity condition remains valid.

B.2.3 Commensurability

IDS used attribute-specific value functions to achieve commensurability by converting different types of evidence into a standardized 0–1 measurement scale. The 82% sentiment-analysis score received a linear transformation to produce a value of 0.82. The Licence & Regulation category received separate evaluation through five distinct sub-attributes, namely: (i) Commercial use, (ii) Derivatives & redistribution, (iii) Licence type, (iv) Usage restrictions, and (v) Documentation clarity. The sub-attributes received scores from 0 to 100, which were combined through branch weights that placed more importance on Commercial use and Licence type before normalization to $[0,1]$ for the additive value model.

Appendix C – Evaluation Data

C.1 Raw Benchmark Scores

	FinMa	FinLLaMA	Llama-2	Mistral-7B	FinTral
SA	0.88	0.82	0.58	0.49	0.81
NER	0.64	0.57	0.02	0	0.4
NU	0.12	0.69	0	0	0.02
TS	0.07	0.37	0.3	0.3	0.4
SMP	0.51	0.56	0.5	0.49	0.53
CS	0.29	0.56	0.52	0.48	0.61
Context Window	2048	8000	4000	8192	8000

Table C.1: Raw, unscaled benchmark values for all six sub-criteria of performance and one sub-criterion of operational capability before linear transformation.

C.2 Licence Assessment Matrix

Model	A: Commercial use	B: Derivatives & redistribution	C: License type	D: Usage restrictions	E: Documentation clarity	Score /100
FinMA-7B	100	100	100	95	90	98.5
Mistral-7B	100	100	100	95	90	98.5
Llama-2	85	70	60	60	85	73.8
FinLLaMA	85	70	60	60	80	73.3
FinTral	20	10	20	40	20	19.5

Table C.2: Licence Assessment Matrix

C.3 Ranking of Alternatives

C.3.1 Performance

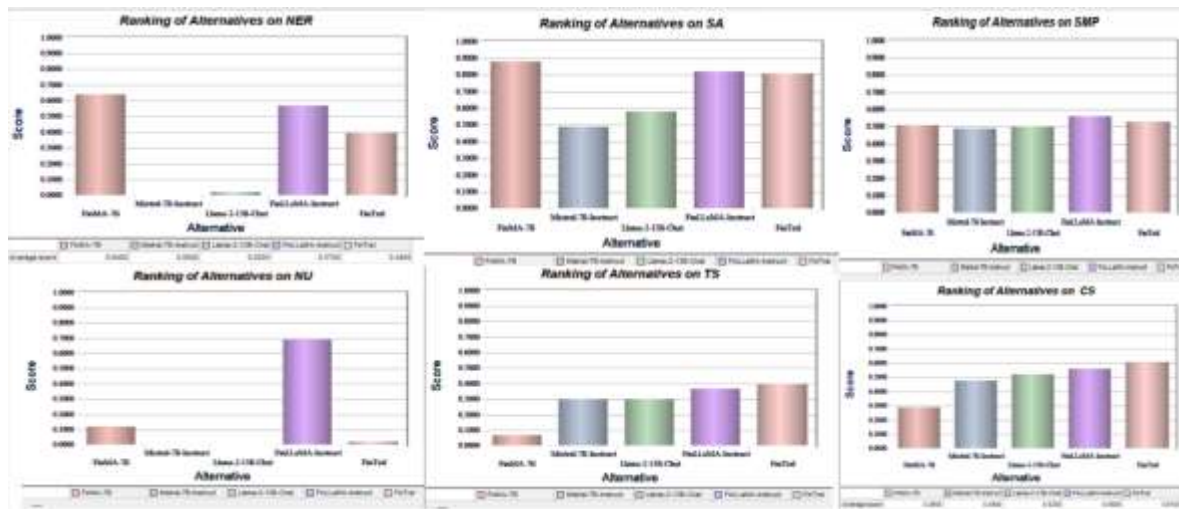


Figure C.3.1: Ranking of Alternatives on sub-criteria of Performance

C.3.2 Operational Capability

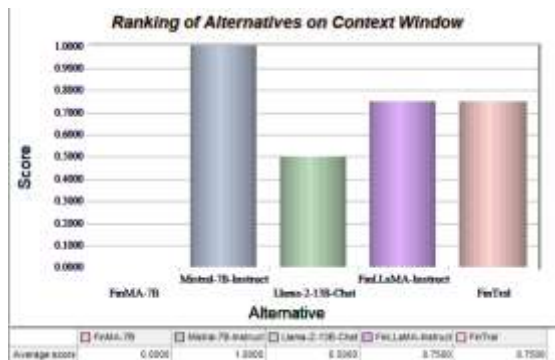


Figure C.3.2: Ranking of Alternatives on sub-criteria of Operational Capability

C.3.3 Licence & Regulation

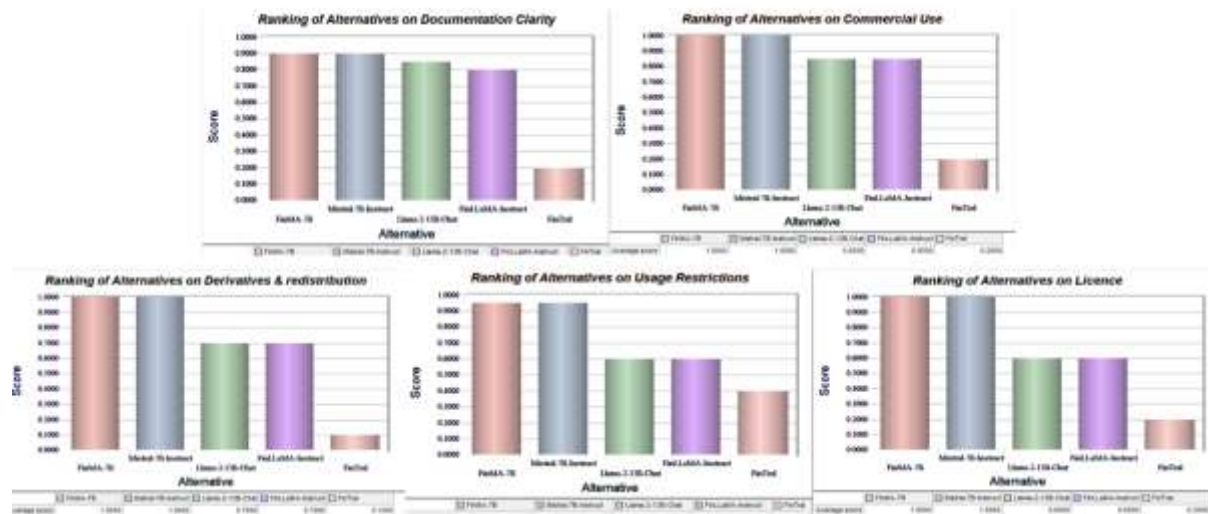


Figure C.3.3 Ranking of Alternatives on sub-criteria of Licence and Regulation